



POWER BI MAJOR CAPSTONE PROJECT

## END TO END POWER BI PROJECT

---

### PROJECT TITLE -                      TECH WORKFORCE SALARY & FINANCIAL ANALYTICS DASHBOARD

---

PROJECT CATEGORY              HR Analytics  
Salary Analytics  
Financial Analytics  
Workforce Analytics

---

INSTITUTE NAME                      Infinity Code Nexus

---

TRAINER / GUIDE                      Anand Kumar Anamaina

---

CREATED BY                              KANAMATHAREDDY CHETHAN REDDY

---

TOOLS USED                              Power BI Desktop Power Query, DAX & Power BI  
Service

---

SUBMISSION DATE                      [14/06/2026]

---

---

Prepared as part of the Power BI Masterclass Program to demonstrate skills in Data Analysis, Data Modeling, DAX Development, Dashboard Design, UI/UX, Storytelling, and Business Intelligence Reporting.

---

## PROJECT OVERVIEW

### PROJECT OBJECTIVE

The objective of this project is to develop an interactive Power BI dashboard solution for analyzing workforce compensation, employee demographics, skills, certifications, benefits, executive performance indicators, and geographical workforce distribution within the technology industry.

The project focuses on transforming raw workforce data into meaningful business insights through data modeling, DAX calculations, KPI tracking, interactive visualizations, drill-down analysis, drill-through navigation, conditional formatting, and custom tooltips.

---

The project successfully delivers insights through the following analytical areas:

- ✓ Tech Salary Analytics Dashboard
- ✓ Salary Analysis Dashboard
- ✓ Workforce & Demographics Analytics Dashboard
- ✓ Skills & Certification Analytics Dashboard
- ✓ Compensation & Benefits Analytics Dashboard
- ✓ Executive Insights Dashboard
- ✓ Global Location & Regional Insights Dashboard

The dashboard solution enables organizations to evaluate salary trends, workforce composition, employee satisfaction, compensation structures, skill development, regional workforce distribution, and strategic business performance using data-driven decision-making.

## **PROJECT GOAL**

The goal of this project is to transform workforce, compensation, demographic, skills, certification, and geographic data into interactive business intelligence dashboards that support data-driven decision-making.

The Power BI solution provides a comprehensive view of employee compensation, workforce demographics, career growth, certification impact, compensation benefits, executive performance indicators, and regional workforce distribution. Through KPI monitoring, advanced visualizations, drill-down analysis, drill-through navigation, custom tooltips, and conditional formatting, the project enables stakeholders to identify compensation trends, workforce patterns, high-performing technology domains, and strategic opportunities for organizational growth.

The dashboard helps organizations improve workforce planning, compensation management, employee engagement, talent development, and executive decision-making through actionable insights derived from real-world workforce data.

---

## **BUSINESS PROBLEM**

Technology organizations generate large volumes of workforce, compensation, and employee performance data; however, extracting meaningful insights from this information remains a challenge. Decision-makers often struggle to understand how salary structures, experience levels, education, certifications, company size, work arrangements, and employee satisfaction influence workforce performance and organizational growth.

Without a centralized analytical solution, identifying compensation trends, workforce demographics, skill development patterns, and regional workforce distribution becomes time-consuming and inefficient. This limits an organization's ability to make informed decisions regarding talent acquisition, employee retention, compensation planning, and workforce optimization.

To address these challenges, this Power BI solution was developed to provide comprehensive insights into salary analytics, workforce demographics, skills and certifications, compensation and benefits, executive workforce performance, and global workforce distribution through interactive dashboards and KPI-driven reporting.

The project aims to answer critical business questions such as:

- Which technology domains and job roles offer the highest compensation?
- How does experience influence employee salary growth?
- What impact do certifications and education levels have on compensation?
- How do company size and employment type affect salary and benefits?
- Which workforce groups demonstrate higher job satisfaction levels?
- How are employees distributed across different geographic regions?
- What factors contribute to workforce engagement and career growth?
- How can organizations use workforce analytics to support strategic decision-making?

## **BUSINESS QUESTIONS**

The Power BI dashboard solution was developed to analyze and answer the following business questions:

1. What is the average annual salary, net salary, bonus, tax, and savings of employees across the technology industry?
2. Which technology domains offer the highest compensation and career growth opportunities?
3. Which job roles receive the highest salaries and compensation packages?
4. How does employee experience influence salary growth and workforce compensation trends?
5. What impact do certifications, skills, and educational qualifications have on employee earnings?
6. How do company size, employment type, and work arrangements affect compensation and benefits?
7. Which workforce groups demonstrate higher job satisfaction and employee engagement levels?
8. How are employees distributed across gender, age groups, education levels, and geographic locations?
9. Which regions and locations offer the highest average salaries and

compensation opportunities?

10. What workforce patterns and compensation trends can support executive-level decision-making and strategic planning?

11. Which employee groups are more likely to actively seek new job opportunities?

How can workforce analytics help organizations improve talent management, employee retention, and business performance?

12. What is the average annual salary, net salary, bonus, tax, and savings of employees across the technology industry?

13. Which technology domains offer the highest compensation and career growth opportunities?

14. Which job roles receive the highest salaries and compensation packages?

15. How does employee experience influence salary growth and workforce compensation trends?

16. What impact do certifications, skills, and educational qualifications have on employee earnings?

17. How do company size, employment type, and work arrangements affect compensation and benefits?

18. Which workforce groups demonstrate higher job satisfaction and employee engagement levels?

19. How are employees distributed across gender, age groups, education levels, and geographic locations?

20. Which regions and locations offer the highest average salaries and compensation opportunities?

21. What workforce patterns and compensation trends can support executive-level decision-making and strategic planning?

22. Which employee groups are more likely to actively seek new job opportunities?

23. How can workforce analytics help organizations improve talent management, employee retention, and business performance?

# DATASET INFORMATION

---

DATASET NAME : Tech Salary Dataset

DATASET DETAILS : Approximate Records 10,000+

Columns : 50

## EXPECTED ANALYSIS AREAS

---

The project focuses on the following analytical areas:

- Tech Salary Analytics
- Salary & Compensation Analytics
- Workforce & Demographic Analytics
- Skills & Certification Analytics
- Compensation & Benefits Analytics
- Executive Workforce Insights
- Global Location & Regional Analytics
- Employee Satisfaction Analytics
- Career Growth & Experience Analytics
- Workforce Distribution Analytics
- Geographic Salary Analytics
- Organizational & Employment Type Analytics

# POWER QUERY TRANSFORMATIONS

---

## DATACLEANING ACTIVITIES

- ✓ Validated employee and compensation data
- ✓ Checked for duplicate employee records
- ✓ Handled missing and null values
- ✓ Corrected data types for salary, bonus, tax, age, and experience fields
- ✓ Standardized categorical values across technology domains, education levels, and employment types
- ✓ Verified data consistency and quality

---

## DATA PREPARATION ACTIVITIES

- ✓ Renamed columns for better readability
- ✓ Removed unnecessary columns
- ✓ Formatted numerical and text fields
- ✓ Applied data validation checks
- ✓ Optimized dataset structure for reporting and analysis
- ✓ Prepared data for KPI calculations and dashboard visualizations

## CUSTOM COLUMNS CREATED

=====

- ✓ Experience Group
- ✓ Age Group
- ✓ Salary Category
- ✓ Certification Status
- ✓ Organization Scale
- ✓ Workforce Category

## DATA MODELING

=====

### Star Schema

=====

**Purpose:** A Star Schema data model was implemented to optimize report performance, simplify DAX calculations, and improve dashboard scalability. The model was designed around a central employee salary dataset and supporting analytical dimensions.

#### FACT TABLE

tech\_salary\_dataset

Contains employee-level information including:

- Employee ID
- Job Title
- Technology Domain
- Salary Details
- Bonus Information
- Tax Information
- Savings
- Experience
- Certifications
- Education Level
- Job Satisfaction
- Work Arrangement
- Location

## **DIMENSION TABLES**

The following analytical dimensions were used:

- Technology Domain
- Job Title
- Education Level
- Company Size
- Employment Type
- Gender
- Location
- Work Arrangement

## RELATIONSHIPS

- One-to-Many Relationships
- Single Direction Filtering
- Optimized for KPI and Dashboard Analysis

## MODEL BENEFITS

- Faster Dashboard Performance
- Simplified DAX Development
- Better Data Organization
- Improved Scalability
- Consistent Reporting Across Dashboards

## DAX MEASURES USED

---

### EMPLOYEE COUNT

Employee Count =  
DISTINCTCOUNT(tech\_salary\_dataset[employee\_id])

PURPOSE: Calculates the total number of unique employees in the dataset.

### AVG ANNUAL SALARY

Avg Annual Salary =  
AVERAGE(tech\_salary\_dataset[annual\_salary\_usd])

PURPOSE : Calculates the average annual  
salary of employees

### AVG NET SALARY

Avg Net Salary =  
AVERAGE(tech\_salary\_dataset[annual\_net\_salary\_usd])

PURPOSE : Calculates the average annual net salary after deductions.

## **AVG BONUS**

Avg Bonus =  
AVERAGE(tech\_salary\_dataset[annual\_bonus\_usd])

PURPOSE: Calculates the average annual bonus received by employees.

## **AVG TAX**

Avg Tax =  
AVERAGE(tech\_salary\_dataset[annual\_tax\_usd])

PURPOSE: Calculates the average annual tax paid by employees.

## **AVG SAVINGS**

Avg Annual Savings =  
AVERAGE(tech\_salary\_dataset[annual\_savings\_usd])

PURPOSE: Calculates the average annual savings of employees.

## **AVG EXPERIENCE**

Avg Experience =  
AVERAGE(tech\_salary\_dataset[experience\_years\_total])

PURPOSE: Calculates the average years of professional experience.

## **AVG AGE**

Avg Age =  
AVERAGE(tech\_salary\_dataset[age])

PURPOSE: Calculates the average employee age.

## **AVG CERTIFICATIONS**

Avg Certifications =  
AVERAGE(tech\_salary\_dataset[certifications])

PURPOSE: Calculates the average number of certifications held by employees.

## **AVG HOURLY RATE**

Avg Hourly Rate =  
AVERAGE(tech\_salary\_dataset[hourly\_rate\_usd])

PURPOSE: Calculates the average hourly compensation rate.

## **AVG JOB SATISFACTION**

Avg Job Satisfaction =  
AVERAGE(tech\_salary\_dataset[job\_satisfaction\_score])

PURPOSE: Calculates the average employee satisfaction score.

## **ACTIVE JOB SEEKERS %**

Active Job Seekers % =  
DIVIDE(  
CALCULATE(  
COUNTROWS(tech\_salary\_dataset),  
tech\_salary\_dataset[is\_actively\_looking] = TRUE()  
),  
COUNTROWS(tech\_salary\_dataset)  
)

PURPOSE: Calculates the percentage of employees actively looking for new job opportunities.

## **LOCATIONS COVERED**

Locations Covered =  
DISTINCTCOUNT(tech\_salary\_dataset[location])

PURPOSE: Calculates the number of unique employee locations represented in the dataset.

# VISUALIZATIONS USED

---

## **KPI CARDS:**

Used to display:

- Employee Count
- Avg Annual Salary
- Avg Net Salary
- Avg Bonus
- Avg Tax
- Avg Savings
- Avg Experience
- Avg Age
- Avg Job Satisfaction
- Active Job Seekers %

**Bar Charts:** Used to compare:

- Technology Domain Salaries
- Top Paying Job Roles
- Salary by Education Level
- Workforce Distribution

**Line Charts:** Used to compare:

- Salary Trend by Experience
- Skill Ranking by Salary Growth
- Experience Distribution

**Donut Charts:** Used to compare:

- Employment Type Distribution
- Workforce Composition

**Treemap :**

Used to compare salary distribution across company sizes.

**Funnel Chart:**

Used to analyze workforce distribution by education level.

**Waterfall Chart:**

Used to evaluate salary contribution by company size.

**Map Visual:**

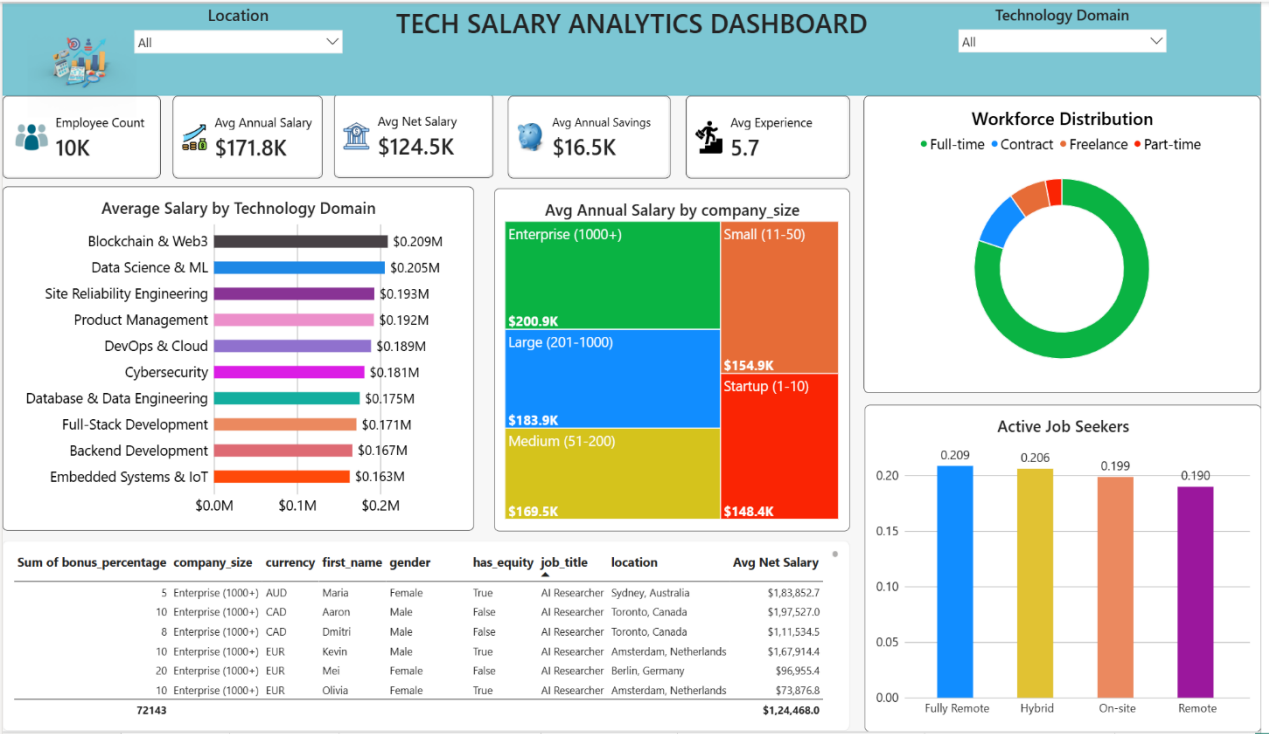
Used to analyze salary distribution across global locations.

**Tables & Matrices:**

Used to display detailed employee records and summarized analytical views.

# DASHBOARD SCREENSHOTS

## Dashboard 1 – Tech Salary Analytics Dashboard



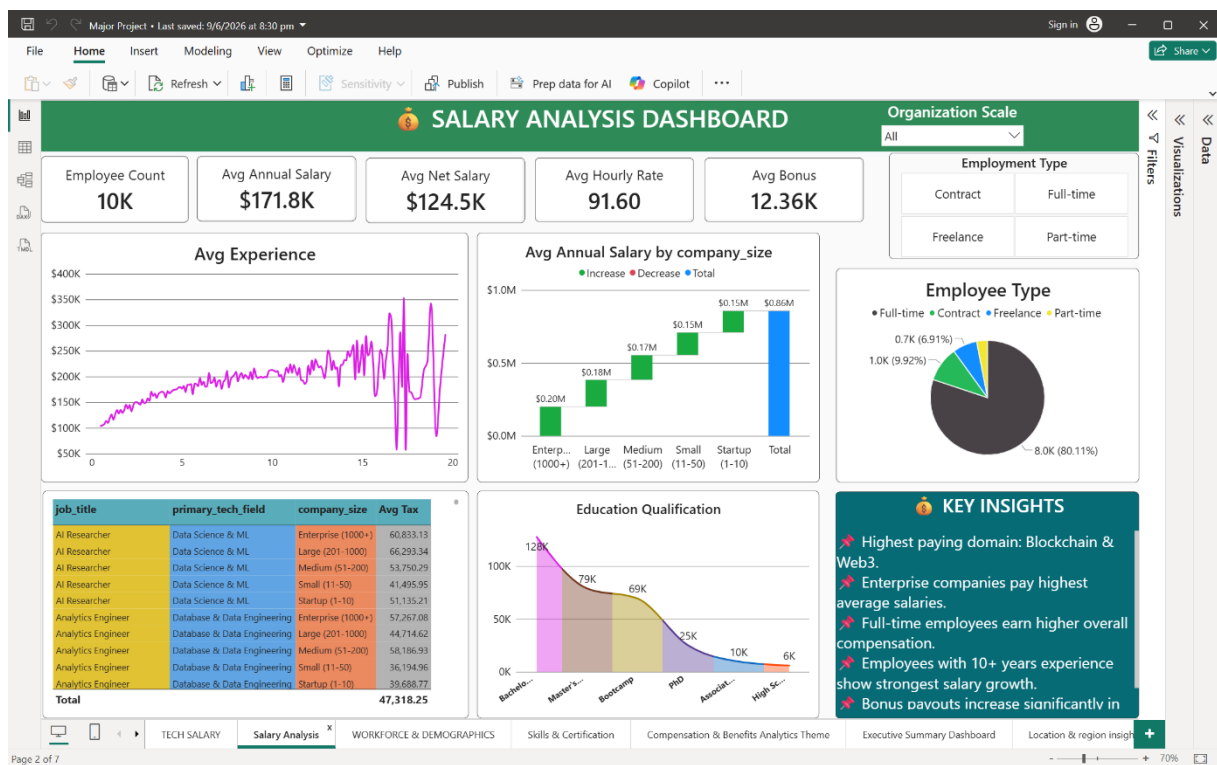
### Explanation :

Provides a high-level overview of workforce compensation, employee count, average salary, net salary, savings, experience, certifications, and job satisfaction. This dashboard serves as the executive summary page for business stakeholders.

### Major KPIs Used:

- Employee Count
- Avg Annual Salary
- Avg Net Salary
- Avg Annual Savings
- Avg Experience

## Dashboard 2 – Salary Analytics Dashboard



### Explanation:

Analyzes salary distribution across technology domains, job roles, company sizes, and education levels. Helps identify the highest-paying sectors and compensation trends.

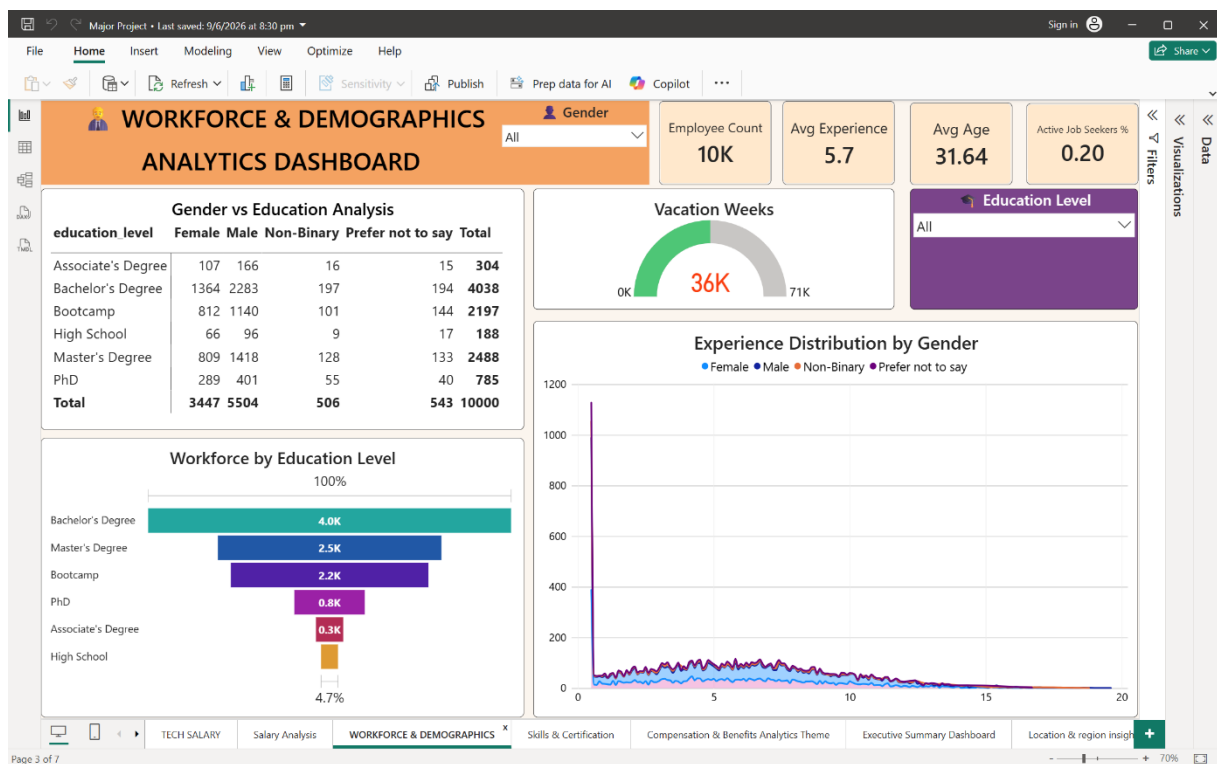
### Major Measures Used:

- Avg Annual Salary
- Avg Bonus
- Avg Net Salary

### Major Visuals:

- Bar Charts
- Treemap
- KPI Cards
- Donut Charts

## Dashboard 3 – Workforce Demographics Analytics Dashboard



### Explanation:

Examines employee demographics such as age, gender, education, work arrangement, and geographical distribution to understand workforce composition.

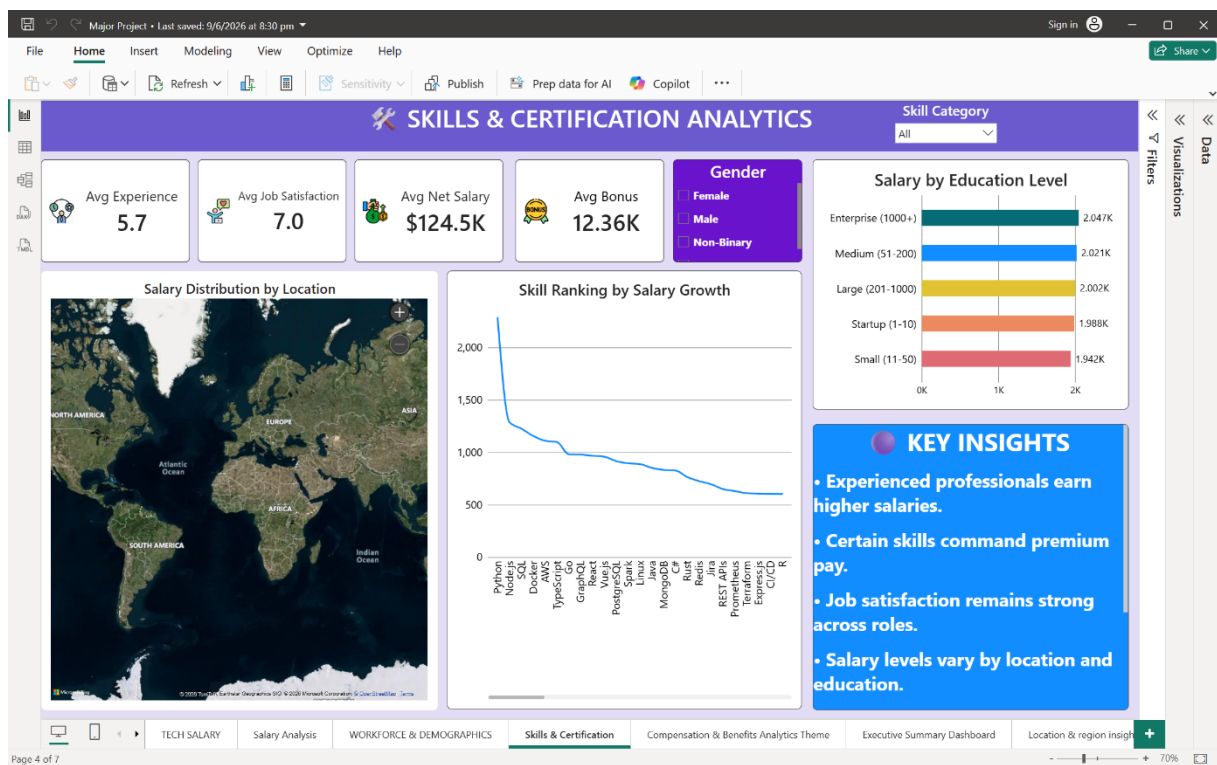
### Major Measures Used:

- Employee Count
- Avg Age
- Avg Experience
- Active Job Seekers %

### Major Visuals:

- Donut Charts
- Funnel Charts
- Maps
- KPI Cards

## Dashboard 4 – Skills Certification Analytics Dashboard



### Explanation:

Analyzes the relationship between experience, education, and salary progression. Highlights career growth patterns within the technology workforce.

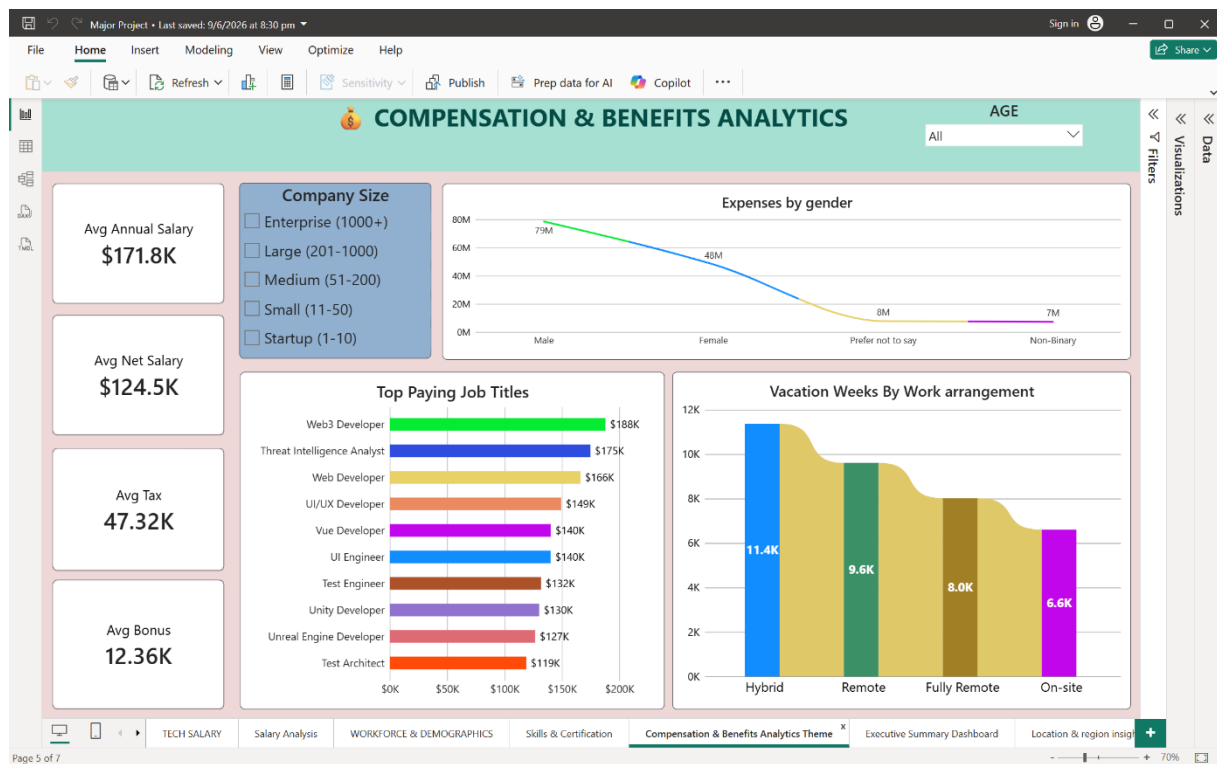
### Major Measures Used:

- Avg Experience
- Avg Job Satisfaction
- Avg Net Salary
- Avg Bonus

### Major Visuals:

- Line Charts
- Bar Charts
- KPI Cards

## Dashboard 5 – Compensation & Benefits Analytics Dashboard



### Explanation:

Focuses on financial metrics including bonuses, taxes, net salary, savings, and compensation structures across different workforce segments.

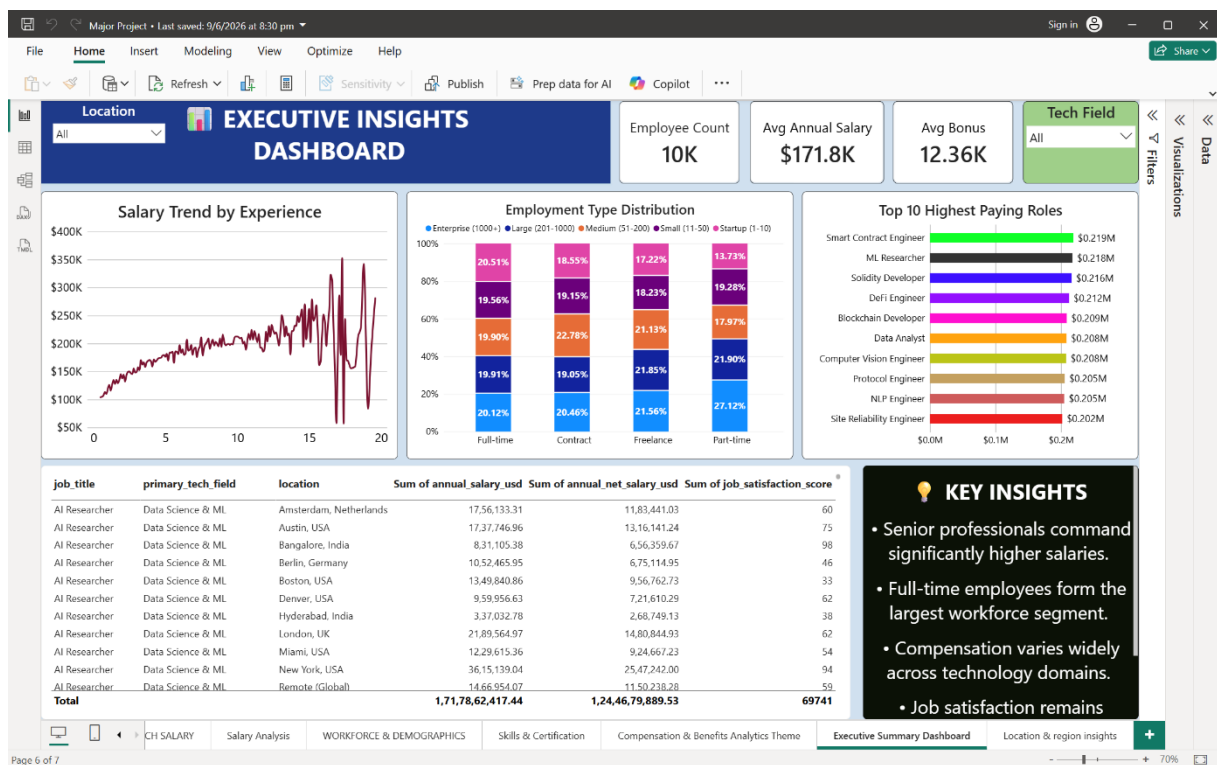
### Major Measures Used:

- Avg Annual Salary
- Avg Net Salary
- Avg Tax
- Avg Bonus

### Major Visuals:

- Waterfall Charts
- Bar Charts
- KPI Cards

## Dashboard 6 – Executive Insights Dashboard



### Explanation:

Provides a high-level summary of workforce compensation, employment distribution, salary trends, and top-paying technology roles. This dashboard enables decision-makers to quickly assess organizational performance, workforce composition, and compensation patterns across technology domains.

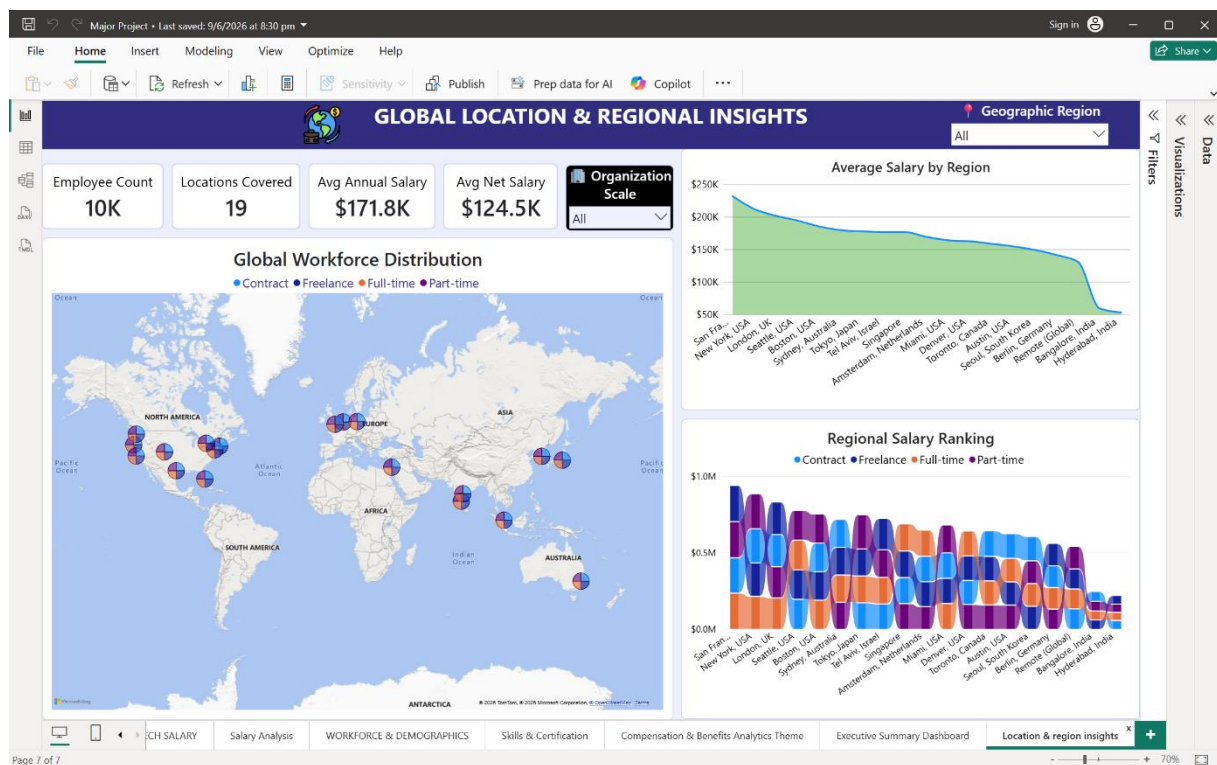
### Major Measures Used:

- Employee Count
- Avg Annual Salary
- Avg Bonus

### Major Visuals:

- KPI Cards
- Line Chart (Salary Trend by Experience)
- 100% Stacked Column Chart (Employment Type Distribution)
- Bar Chart (Top 10 Highest Paying Roles)

## Dashboard 7 – Global Location & Regional Insights Dashboard



### Explanation:

Analyzes workforce distribution across global locations and compares regional salary patterns. This dashboard helps identify high-paying regions, workforce concentration, employment types across countries, and geographic compensation trends.

### Major Measures Used:

- Employee Count
- Locations Covered
- Avg Annual Salary
- Avg Net Salary

### Major Visuals:

- KPI Cards
- Line Chart (Salary Trend by Experience)
- 100% Stacked Column Chart (Employment Type Distribution)
- Bar Chart (Top 10 Highest Paying Roles)
- Data Table
- Insights Panel
- Slicers

# KEY INSIGHTS

---

## KEY INSIGHTS SECTION

After Dashboard 7,

### **Insight 1**

Blockchain & Web3 professionals earn the highest average salaries across technology domains.

### **Insight 2**

Enterprise-sized organizations provide significantly higher compensation packages.

### **Insight 3**

Experienced employees consistently achieve higher salary growth and savings rates.

### **Insight 4**

Certified professionals demonstrate stronger earning potential compared to non-certified employees.

### **Insight 5**

Average employee satisfaction remains above 7/10, indicating a generally positive workforce environment.

### **Insight 6**

A considerable percentage of employees are actively seeking new opportunities, highlighting retention challenges.

### **Insight 7**

Compensation, certifications, and experience are the strongest drivers of workforce earnings.

# BUSINESS RECOMMENDATIONS

---

## RECOMMENDATION-1

Invest in employee training and certification programs to enhance technical expertise and improve earning potential across technology domains.

## RECOMMENDATION-2

Promote continuous upskilling initiatives in emerging technologies such as Artificial Intelligence, Data Science, Cloud Computing, Cybersecurity, and Blockchain.

## RECOMMENDATION-3

Develop employee retention strategies focusing on competitive compensation, career growth opportunities, and improved work-life balance.

## RECOMMENDATION-4

Implement financial wellness programs that help employees improve savings habits, financial planning, and long-term wealth management.

# FEATURES IMPLEMENTED

---

- ✓ KPI Cards
- ✓ Interactive Slicers
- ✓ Drill Through Functionality
- ✓ Cross-Filtering Visuals
- ✓ Custom Dashboard Themes
- ✓ Dynamic Insights Panel
- ✓ Conditional Formatting
- ✓ Geographic Mapping
- ✓ Responsive Dashboard Layout
- ✓ Bookmarks & Navigation
- ✓ Multi-Page Reporting

# CHALLENGES FACED

---

- Handling large workforce datasets and ensuring data quality.
- Creating meaningful DAX measures for business KPIs.
- Designing consistent dashboard themes across multiple report pages.
- Optimizing visual performance and report responsiveness.
- Building effective relationships between multiple data attributes.
- Developing business-focused insights from complex workforce data.

# LEARNINGS

---

- Data Cleaning and Preparation
- Data Modeling and Relationships
- DAX Measure Development
- KPI Design and Dashboard Development
- Workforce Analytics and Business Intelligence
- Interactive Report Design
- Storytelling with Data
- Business Decision-Making through Analytics

# PROJECT STORYTELLING

---

## **BUSINESS SCENARIO**

Technology organizations generate vast amounts of workforce and compensation data related to salaries, experience, skills, certifications, education, job satisfaction, and work arrangements. Decision-makers require an analytical solution that transforms this data into actionable insights for workforce planning, compensation management, and employee development.

## **BUSINESS PROBLEM**

- Which technology domains provide the highest compensation?
- How does professional experience influence salary growth?
- Which job roles receive the highest compensation packages?
- Do certifications and educational qualifications contribute to higher salaries?
- Which workforce segments demonstrate higher job satisfaction?
- How do work arrangements impact compensation and employee engagement?
- Which geographic locations offer better salary opportunities?

## **ANALYSIS APPROACH**

- Analyzed compensation patterns across technology domains and job roles.
- Compared salary trends against experience, education, and certifications.
- Evaluated workforce demographics and employment distribution.
- Assessed employee satisfaction and active job-seeking behavior.
- Examined compensation, benefits, taxes, and savings metrics.
- Analyzed global workforce distribution and regional salary variations.
- Generated executive-level insights to support business decision-making

## KEY FINDINGS

- Blockchain & Web3 professionals earn the highest average salaries.
- Enterprise organizations offer the strongest compensation packages.
- Salary growth increases significantly with professional experience.
- Certified professionals generally demonstrate higher earning potential.
- Full-time employees represent the largest workforce segment.
- Employee satisfaction remains consistently strong across technology sectors.
- Global salary distribution varies significantly across regions and locations.

## BUSINESS IMPACT

- Supports strategic compensation planning.
- Enhances workforce development initiatives.
- Improves talent acquisition and retention strategies.
- Provides actionable insights for executive decision-making.
- Enables better resource allocation and workforce planning.
- Strengthens data-driven organizational management.

## PROJECT DEPLOYMENT

### DEPLOYMENT OBJECTIVE

Deploy the Power BI Tech Salary Analytics Solution to provide stakeholders with real-time access to workforce, compensation, demographic, and financial insights through interactive dashboards.

### DEPLOYMENT PLATFORM

Power BI Service

## **DEPLOYMENT PROCESS**

### **STEP-1**

Validate dataset quality, data model relationships, and DAX measures within Power BI Desktop.

### **STEP-2**

Optimize report performance, dashboard layouts, and visual interactions.

### **STEP-3**

Publish the PBIX report to Power BI Service.

### **STEP-4**

Configure dataset refresh schedules and gateway settings.

### **STEP-5**

Create executive dashboards and reporting views.

### **STEP-6**

Implement role-based security and access permissions.

### **STEP-7**

Share dashboards with stakeholders and business users.

## **BUSINESS BENEFITS**

- Centralized Workforce Analytics Platform
- Interactive Salary & Compensation Monitoring
- Real-Time Business Intelligence Reporting
- Improved Strategic Workforce Planning
- Faster Data-Driven Decision-Making
- Enhanced Executive Reporting Capabilities

## CONCLUSION

---

The Tech Salary Analytics Dashboard successfully transformed workforce, compensation, demographic, skills, and financial data into meaningful business intelligence. Through seven interactive dashboards, the solution provides comprehensive visibility into salary trends, workforce demographics, skills and certifications, compensation structures, executive insights, and global workforce distribution.

The project demonstrates how Power BI can be leveraged to support workforce planning, compensation management, employee development, and strategic business decision-making through data-driven insights and interactive reporting.

---